THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The area of the triangle with vertices $(0,0)$, $(10,0)$ and $(0,14)$ is: 1

(a) 69 **(b)** 72 **(c)** 70 **(d)** 71

2. The area (in square units) of the triangle formed by the line $y = 2x + 10$ and the two axes is: 1

(a) 25 **(b)** 27 **(c)** 24 **(d)** 26

3. The value of $0.\overline{2} + 0.\overline{5}$ is: 1

(a) $\frac{8}{9}$ **(b)** $\frac{7}{9}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{7}{10}$

4. If $x + y = 5$ and $xy = 6$, then $x^3 + y^3 =$ 1

(a) 36 **(b)** 35 **(c)** 37 **(d)** 34

5. Parallel chords of lengths 14 cm and 48 cm lie on opposite sides of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 30 **(b)** 33 **(c)** 32 **(d)** 31

6. The area of a cyclic quadrilateral with sides 5 cm, 7 cm, 15 cm and 19 cm is (cm^2): 1

(a) 98 **(b)** 95 **(c)** 97 **(d)** 96

7. Two dice are rolled together. The probability that the product of the numbers is 12 is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{2}{9}$ **(d)** $\frac{1}{10}$

8. The 6th term of an AP is 20 and the 10th term is 32. The 16th term is: 1

(a) 53 **(b)** 50 **(c)** 47 **(d)** 56

9. The number of perfect squares between 100 and 1000 (both excluded) is: 1

(a) 23 **(b)** 22 **(c)** 21 **(d)** 20

10. A group of 34 people has average weight 111 kg. Another group has average weight 72 kg. Together the average weight is 78 kg. The number of people in the second group is: 1

(a) 189 **(b)** 188 **(c)** 186 **(d)** 187

11. Start with $n = 26$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 11 **(b)** 12 **(c)** 10 **(d)** 9

12. The four angles of a quadrilateral are in AP with common difference 50°. The smallest angle is: 1

(a) 15° **(b)** 25° **(c)** 35° **(d)** 65°

13. The line $17x + 18y = 306$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 154 **(b)** 155 **(c)** 152 **(d)** 153

14. The diagonal of a cuboid measuring 12 cm $\times$ 24 cm $\times$ 24 cm is: 1

(a) 38 cm **(b)** 35 cm **(c)** 37 cm **(d)** 36 cm

15. The distance of the circumcentre of the right triangle with vertices $(0, 0)$, $(6, 0)$ and $(0, 8)$ from the origin is: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 6

16. The x-coordinate of the point where $y = 5x + 1$ and $y = 2x + 13$ meet is: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

17. **Assertion (A):** The point $(-3, 4)$ lies in Quadrant II. 1

Reason (R): In Quadrant II the abscissa is negative and the ordinate is positive.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** The zero of the polynomial $p(x) = 3x - 12$ is 4. 1

Reason (R): The zero of $ax + b$ $(a \neq 0)$ is $-\dfrac{b}{a}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** The number 0.375 is a rational number. 1

Reason (R): Every terminating decimal can be written in the form $\dfrac{p}{q}$ with integers p, q and $q \neq 0$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** $103^2 = 10609$. 1

Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. The vertices of a right triangle are $O(0,0)$, $A(6,0)$ and $B(0,8)$. Find the circumcentre and the circumradius. **2**

22. Plan A costs ₹200 plus ₹30 per unit; Plan B costs ₹440 plus ₹10 per unit. For how many units do both cost the same? **2**

23. Express $0.\overline{17} + 0.\overline{71}$ as a single fraction in lowest terms. **2**

24. If $x + y = 8$ and $xy = 12$, find $x^3 + y^3$. **2**

25. Two parallel chords of lengths 14 cm and 48 cm lie on the opposite sides of the centre of a circle of radius 25 cm. Find the distance between them. **2**

26. Find the area of a cyclic quadrilateral with sides 5, 7, 15 and 19 cm. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Find the points on the x-axis whose distance from $A(2,20)$ is 25 units. **3**

28. Find the point of intersection of $y = 5x + 2$ and $y = 2x + 11$. **3**

29. Express $0.4\overline{7}$ as a fraction in lowest terms and verify by long division. **3**

30. If $x + \frac{1}{x} = 7$, find $x^3 + \frac{1}{x^3}$. **3**

31. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm. Find the distance between their centres if they are on
(i) opposite sides,
(ii) the same side of the common chord. **3**

OR

A rope 44 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. **3**

32. A bag contains red, blue and green balls. $P(\text{red}) = \frac{1}{3}$ and $P(\text{blue}) = \frac{1}{3}$ and there are 24 balls in all. Find $P(\text{green})$ and the number of balls of each colour. **3**

OR

The sum of the 3rd and 9th terms of an AP is 40 and its 5th term is 17. Find the AP's first term and the sum of its first 20 terms. **3**

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** On the map of a town drawn on a grid (1 unit = 1 km), Kavya's school is at A $(-3,4)$, a park is at B$(-3,-2)$ and the market is at C$(5,-2)$. **5**
(i) In which quadrant does the school lie?
(ii) Write the coordinates of the reflection of the school in the x-axis.
(iii) Find the distance between the school and the park.
(iv) A bus stop is to be placed midway between the school A and the market C. Find its coordinates and the distance AC.

34. **Case study.** A taxi company in Jaipur charges a fixed amount of Rs 50 plus Rs 12 for every kilometre travelled. The fare for x km is given by a polynomial $C(x)$. 5

(i) Write $C(x)$.

(ii) Find the fare for a 10 km ride.

(iii) For what distance is the fare Rs 230?

(iv) State the slope and the y-intercept of the graph of $y = C(x)$ and say what each means.

35. Find the decimal expansion of $\frac{3}{7}$ by long division, state the repeating block and its length, and hence write $\frac{3}{7}$ as a repeating decimal. 5

36. If $x - y = 3$ and $xy = 28$, find $x^2 + y^2$, $x^3 - y^3$ and $(x + y)^2$. 5

OR

In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 10)^\circ$, $\angle B = (4y + 8)^\circ$, $\angle C = (3x + 20)^\circ$ and $\angle D = (2y + 4)^\circ$. Find x, y and all four angles. 5

37. In quadrilateral $ABCD$, $AB = 6$ cm, $BC = 8$ cm, $CD = 17$ cm, $DA = 21$ cm and $\angle ABC = 90^\circ$. Find the area of $ABCD$. 5

OR

Two dice are rolled together. Find the probability that 5

(i) the sum is 6,

(ii) the sum is more than 6,

(iii) the sum is at most 6. Verify that

(ii) and

(iii) add to 1.

38. A contractor pays a penalty for delay: ₹200 on day 1, ₹250 on day 2, ₹300 on day 3 and so on. If the work is 30 days late, find the penalty on the last day and the total penalty. 5

OR

Prove that the product of three consecutive integers is divisible by 6. Verify it for $5, 6, 7$. 5

END OF QUESTION PAPER **Total: 80 Marks**

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